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Inventor(s)	Zhang; Liang et al.

Voltage regulator with on-time extension

Abstract

A voltage regulator circuit includes a high side (HS) transistor having a control input and a low side (LS) transistor having a control input. The LS transistor is coupled to the HS transistor at a switching terminal. A comparator has first and second inputs and an output. A first resistor is coupled to the switching terminal. A second resistor is coupled between the first resistor and the second input of the comparator. A capacitor is coupled between a second resistor terminal of the second resistor and ground. A switch has first and second switch terminals and a control input. The first switch terminal is coupled to the first resistor terminal of the second resistor, and the second switch terminal is coupled to the second resistor terminal. A delay circuit has an input and an output. The output of the delay circuit is coupled to the control input of the switch.

Inventors: Zhang; Liang (Beijing, CN), Wu; Guoyang (Beijing, CN), Jing; Weibing (Beijing, CN), Li; Dan (Beijing, CN)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

Family ID: 76317343

Assignee: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

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Primary Examiner: Novak; Peter M

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 17/110,504 filed Dec. 3, 2020, which claims priority to U.S. Provisional Application 62/948,040, filed Dec. 13, 2019, titled “DCDC Converter with On-Time Extension (OTE) Circuit,” both of which are hereby incorporated by reference in their entirety.

BACKGROUND

(1) A direct current (DC)-to-DC switching voltage regulator may include a high side (HS) transistor coupled to a low side (LS) transistor. The DC-to-DC voltage regulator may also include a control circuit that turns on and off the HS and LS transistors to produce a target output voltage on the output of the regulator. Various control techniques can be employed to control the operation of the HS and LS transistors. In one example, the control technique implements a fixed on-time control for the HS transistor—so in each cycle, the HS transistor is turned on for a fixed period of time. To regulate the level of the output voltage for a range of possible input voltages, the off-time of the HS transistor (during which the LS transistor is turned on) can be varied, thereby varying the duty cycle (i.e., the percentage of the time that the HS transistor is on in each cycle). At lower levels of input voltage (which may occur if the input voltage is supplied by a battery whose charge is depleting), a fixed on-time control circuit responds by increasing the duty cycle.

SUMMARY

(2) In one example, a voltage regulator circuit includes a high side (HS) transistor having a control input and a low side (LS) transistor having a control input. The LS transistor is coupled to the HS transistor at a switch terminal. A comparator has first and second inputs and an output. A first resistor is coupled to the switch terminal. A second resistor is coupled between the first resistor and the second input of the comparator. A capacitor is coupled between a second resistor terminal of the second resistor and ground. A switch has first and second switch terminals and a control input. The first switch terminal is coupled to the first resistor terminal of the second resistor, and the second switch terminal is coupled to the second resistor terminal. A delay circuit has an input and an output. The output of the delay circuit is coupled to the control input of the switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a voltage regulator employing on-time extension for a high side transistor.
- (2) FIG. 2 shows an example implementation of a delay circuit in the voltage regulator.
- (3) FIG. 3 shows a timing diagram illustrating the operation of the delay circuit of FIG. 2.
- (4) FIG. 4 illustrates output voltage undershoot of a voltage regulator.
- (5) FIGS. 5 and 6 show timing diagrams illustrating the operation of the voltage regulator without on-time extension (FIG. 5) and with on-time extension (FIG. 6).
- (6) FIG. 7 illustrates a reduction in output voltage undershoot of the voltage regulator of FIG. 1.
- (7) FIG. 8 illustrates a system in which the voltage regulator of FIG. 1 can be used.

DETAILED DESCRIPTION

(8) For a fixed on-time control technique for a switching voltage regulator, the duty cycle increases as described above responsive to decreases in the input voltage. The on-time of the HS transistor remains constant, but the off-time decreases thereby resulting in a higher duty cycle. However, there is an upper limit on the duty cycle, and further accommodations to even lower input voltages are not possible. Further, sudden increases in load (e.g., a few milliamperes (mA) of load current jumping to 6 amperes (A) of current when the input voltage is near the output voltage) can result in large drop in the output voltage (referred to as “undershoot”). The examples described herein include a control circuit for a switching DC-to-DC voltage regulator that includes an on-time extension circuit which lengths the otherwise fixed on-time for the HS transistor responsive to periods of time characterized by levels of output voltage undershoot in excess of a threshold amount. By extending the on-time, the load current is able to step up from a low level to a higher level in fewer switching cycles than would otherwise have been the case if the on-time was not extended. As a result, the magnitude of the undershoot is reduced. Further, the duty cycle can achieve a higher level than if the on-time extension capability was absent.

(9) FIG. 1 shows an example of a voltage regulator **100** that includes a control circuit **110** coupled to a high side (HS) transistor and a low side (LS) transistor. The HS and LS transistors in this example are n-type metal oxide semiconductor field effect transistors (NMOS). The drain of the HS transistor is connected to the supply voltage rail (VIN), and the source of the HS transistor is coupled to the drain of the LS transistor. The connection between the HS and LS transistors is referred to as the switching node (SW). The source of the LS transistor is connected to ground. SW is connected to one terminal of inductor L, and the other terminal of inductor L is connected to capacitor C1 and an output resistor R0 at an output node identified as VOUT (VOUT also refers to the output voltage on the node). The logic **170** of the control circuit **110** controls the on and off state of the HS and LS transistors so as to regulate the output voltage (VOUT) to a prescribed level despite the input voltage (VIN) being any voltage within a range of possible input voltages. For example, VOUT may be 5V while VIN is a voltage in the range of 7.4V to 8.4V. When logic **170**

turns on the HS transistor (with the LS transistor being off), the voltage on SW is approximately equal to VIN (VIN less the drain-to-source voltage drop of the HS transistor). When logic **170** turns the LS transistor on (and the HS transistor is off), the voltage on SW is approximately equal to ground. Thus, the voltage on SW toggles between VIN and ground.

(10) A resistor divider comprising resistors R5 and R6 produces a scaled version of VOUT, which is a feedback voltage (VFB) provided to the control circuit **110**. The control circuit **110** includes a ripple injection circuit **120**, a summing comparator **130**, a one-shot circuit **150**, and logic **170**. The ripple injection circuit **120** includes resistors R1-R4, and capacitors C2 and C3. Resistor R1 is connected to SW, and thus receives VIN when the HS transistor is on. R2 is connected between R1 and ground. R3 is connected to R1 and R2 as shown. R3 and R4 are connected together at a node labeled CSP. C2 is connected between CSP and ground. R4 and C3 are connected in series between CSP and ground, with the node between R4 and C3 designated as CSN.

(11) Ripple injection is implemented by the ripple injection circuit and is used for loop compensation (CSP and CSN) and the generation of the TON_REF signal on CSN to control the on-time of the HS transistor. The combination of R4 and C3 implements a low-pass filter. Per the low-pass filtering function performed by R4 and C3, TON_REF is a direct current (DC) signal whose voltage level is a scaled version of the average signal level on the SW node when the HS transistor is on. The voltage on the SW node is approximately equal to VIN and thus TON_REF is scaled approximation of VIN. $TON_REF = (VIN/N) * \text{Duty Cycle}$, which equals $VOUT/N$, where $N = (R1+R2)/R2$.

(12) The one-shot circuit **150** includes a comparator **155**, resistor RF1, capacitor CF, transistor M1, and on-time extension (OTE) circuit **165**. OTE circuit **165** includes resistor RF2, switch SW1 (which may comprise a transistor), and delay circuit **160**. RF2, switch SW1, and delay circuit RF1, RF2, and CF are connected in series between SW and ground. The terminals of RF2 are labeled **168** and **169**. Switch SW1 connects to terminals **168** and **169** of RF2. Accordingly, RF2 is shorted by SW1 when SW1 is closed (on). The delay circuit **160** has an output that is coupled to and provides a control signal to control the open/closed state of switch SW1. M1 is connected across capacitor CF with the drain of CF connected to terminal **169** and the source of M1 connected to ground. Terminal **169** (node between capacitor CF and resistor RF2) is connected to the negative input of comparator **155**. The signal on terminal **169** is labeled VCAP. The positive input of comparator **155** is connected to CSN and thus receives TON_REF. The output of comparator **155** provides a signal labeled ONESHOT **157**, which is provided to a control input **171** of logic **170**.

(13) The summing comparator **130** includes two pairs of inputs, each pair including a positive input and a negative input. The feedback signal, VFB, is provided to positive input **131**. A reference voltage, VREF, is provided to negative input **132**. CSP and CSN are provided to inputs **133** and **134**, respectively. The summing comparator **130** generates an output signal labeled LOOPRAW. LOOPRAW is asserted high when $VFB + CSN$ is higher than $VREF + CSN$ and is forced low when VFB is lower than VREF. LOOPRAW is provided to another control input **173** of logic **170**.

(14) As will be described below, logic **170** control the on and off state of the HS and LS transistors based on its control input signals, LOOPRAW **135** and ONESHOT **157**. In one example, logic **170** is a digital circuit that is synthesized using a synthesization tool based on the functionality described herein.

(15) FIG. 2 shows an example implementation of delay circuit **160**. In this example, delay circuit **160** includes transistors M2 and M3, resistors R5 and R6, capacitor C4, and an inverter **210**. M2 is a p-type metal oxide semiconductor field effect transistor (PMOS) and M3 is an NMOS transistor. The source of M2 is connected to a supply voltage rail (AVDD). Resistors R5 and R6 are connected in series between the drains of M2 and M3. The source of M3 is connected to ground. The gates of M2 and M3 are connected together and receive LOOPRAW. The drain of M3 is connected to the input of inverter **210**. The inverter's output generates a signal called LOOPRAW_DLY **161**, which as shown in FIG. 1 is used to control switch SW1. In one example, switch SW1 is on (closed)

responsive to LOOPRAW_DLY **161** being logic high and is off (open) responsive to LOOPRAW_DLY being logic low. Capacitor C4 is connected across the drain and source of M3 as shown.

(16) FIG. **3** provides a timing diagram illustrating the operation of delay circuit **160**. The input signal to the delay circuit **160** is LOOPRAW, and the output signal is LOOPRAW_DLY. The output voltage VOUT from the voltage regulator **100** increases when the HS transistor is on and decreases when the LS transistor is on. LOOPRAW has a negative pulse as shown at **301** and **311** when VFB (scaled version of VOUT) is smaller than VREF, which is when VOUT falls to the point at which the HS transistor should be turned on to force VOUT up to maintain a regulated level (with a permitted amount of ripple due to the HS and LS transistors being reciprocally turned on and off). (17) In the example of FIG. **3**, the pulse width of negative pulse **301** is larger than the pulse width of negative pulse **311**, so PW1 is larger than PW2. At **302**, LOOPRAW is high, which causes NMOS M3 to be on and PMOS M2 to be off. With M3 on, INV_IN is pulled low via M3 as shown at **322**, and thus LOOPRAW_DLY (output of inverter **210**) is logic high as shown at **332**. When LOOPRAW falls to logic low at falling edge **303**, M3 turns off and M2 turns on. At that point (M2 on and M3 off), the series connection of resistors R4 and R5 and capacitor C4 are connected between voltage AVDD and ground. INV_IN represents the voltage across capacitor C4. At **322**, the capacitor voltage is 0V and the capacitor begins to charge responsive to M2 being turn on. As a result, INV_IN ramps up as shown at **323**.

(18) The rate at which INV_IN ramps up (the slope) is a function of the combined series resistance of R5 and R6 as well as the capacitance of C4. The charge current (I_{C4}) to capacitor C4 is $AVDD/(R5+R6)$. The smaller is the sum of R5 and R6, the larger is the charge current I_{C4} to capacitor C4. All else being equal, a smaller value of $R5+R6$ results in a larger value of I_{C4} and thus a faster charge rate (slope of **323** is greater), and vice versa. Also, all else being equal, the charge rate is inversely related to the capacitance of C4. Accordingly, a smaller capacitance value of C4 results in a faster charge rate, and vice versa. Moreover, the rate of charge of capacitor C4 can be controlled by selection of the resistances of R5 and R6 and the capacitance of C4. While two resistors R5 and R6 are shown in series in the example of FIG. **3**, in another example, a single resistor can be included, more than two resistors can be connected in series, or other series and/or parallel combinations of resistors can be included to obtain a desired effective resistance between the drain of M2 and capacitor C4. Similarly, series and/or parallel combinations of capacitors can be implemented in place of C4 to obtain a desired capacitance.

(19) As INV_IN ramps at **323**, when INV_IN exceeds the threshold voltage of inverter **210**, the output signal LOOPRAW_DLY will change state from high to low. Thus, falling edge **333** of LOOPRAW_DLY occurs after falling edge **303** of LOOPRAW. The amount of time delay (DELAY) is a function the resistance $R5+R6$ and the capacitance of C4. In one implementation, the circuit components (R5, R6, and C4) are selected so that DELAY equals approximately 200 nanoseconds (ns).

(20) LOOPRAW returns to its logic high level at rising edge **304**. With LOOPRAW high, M2 turns off and M3 turns on. With M3 on, capacitor C4 discharges rapidly through M3 to ground, and thus INV_IN has a similar falling edge **324**, and LOOPRAW_DLY has a corresponding rising edge **334**. Thus, the falling edge **333** of LOOPRAW_DLY is delayed relative to the falling edge **303** of LOOPRAW, but the rising edge **334** of LOOPRAW_DLY is coincident with the rising edge **304** of LOOPRAW.

(21) The right-hand side of FIG. **3** illustrates the operation of the delay circuit **160** when the width PW2 of a negative pulse **311** of LOOPRAW is smaller than the delay implemented by the delay circuit (e.g., 200 ns). INV_IN begins to ramp up at **327** as described above, but the width of the negative pulse **311** is small enough that the input voltage to inverter **210** (INV_IN) does not have a chance to exceed the inverter's threshold voltage. Accordingly, LOOPRAW_DLY does not experience a negative pulse.

(22) FIG. **4** illustrates a problem that can occur when there is a sudden change in load current while

VIN is near VOUT. For example, VIN may normally be in the range of 7.4V to 8.4V but can drop below 7.4V as the battery that provides VIN to power the system in which voltage regulator **100** resides begins to completely discharge. For example, FIG. **4** illustrates the operation of the voltage regulator **100** (absent OTE circuit **165**) for VOUT equal to 5V, VIN equal to 6V, and a sudden load current (IL) change as shown at **401**. The load current increases, in this example, from approximately 0 A to 6 A at a rate of approximately 1.2 A/microsecond. The ripples shown for IL reflect the linear increases in IL as the HS transistor is turned on each cycle. The period of time that the HS transistor is on in each cycle is constant (again, absent the use of OTE circuit **165**). VOUT is regulated by controlling the period of time that the HS transistor is off. Accordingly, logic **170** turns the HS transistor on for a fixed amount of time and then off for a variable amount of time so as to maintain VOUT at a target voltage level (e.g., 5V).

(23) FIG. **4** shows that, with VIN being near VOUT, a sudden change in load current causes VOUT to experience a large undershoot at **405** before VOUT recovers back to its regulated output level. The peak magnitude of the undershoot is shown at **410**. In this example, the undershoot peak at **410** is 740 mV. Such a drop in VOUT that large may be problematic to the correction operation of the load. The inclusion of the OTE circuit **165** helps to decrease the magnitude of the undershoot

(24) FIG. **5** illustrates the operation of the voltage regulator **100** including the OTE circuit **165** when there is no significant undershoot problem (as otherwise is shown in FIG. **4**). The fluctuation in VFB relative to VREF is shown. While VFB is below VREF, LOOPRAW is logic low. Thus, LOOPRAW has negative pulses **510**, which are defined by falling edges **511** and rising edges **512**. The width of the negative pulses is equal to the period of time that LOOPRAW is below VREF. In this example, there is relatively little undershoot in VOUT (and thus VFB), and thus the negative pulses on LOOPRAW are relatively short in duration.

(25) FIG. **5** also illustrates VCAP relative to TON_REF, the two signals provided to the inputs of comparator **155**. VCAP ramps up as shown when the HS transistor is on. ONSHOT **157** is high when TON_REF is larger than VCAP, and low otherwise. Accordingly, ONSHOT **157** has negative pulses **530** (defined by falling edges **531** and rising edges **532**) whose widths are equal to the time period in which VCAP is greater than TON_REF. The control inputs **171** and **173** of logic **170** receive ONSHOT **157** and LOOPRAW **135**, respectively. Logic **170** controls the gate signal to the HS transistor to turn on the HS transistor upon occurrence of falling edges **511**, and to turn off the HS transistor upon occurrence of falling edges **531** of ONSHOT **157**. Logic **170** also asserts a control signal **172** (FIG. **1**) to turn on M1 upon turning the HS transistor off to discharge capacitor C5 before the next on-cycle of the HS transistor starts. The on-time (TON) of the HS transistor (i.e., the period that the HS transistor is on in a cycle) is defined by the time period between each falling edge **511** of LOOPRAW and the subsequent falling edge **531** of ONSHOT.

(26) LOOPRAW_DLY remains high during each cycle because the pulse width of each negative pulse **510** of LOOPRAW is too small for INV_IN to become high enough to reach the threshold voltage of inverter **210** as described above. Switch SW1 in FIG. **1** is on during each cycle because LOOPRAW_DLY remains high. With switch SW1 on, resistor RF2 is bypassed and the charge current to capacitor CF is larger than would be the case if switch SW1 were off and the charge current would also have to flow through resistor RF2.

(27) FIG. **6** illustrates the operation of the voltage regulator **100** when there is a significant undershoot problem. The pulse width of negative pulse **610** (defined by falling edge **611** and rising edge **612**) of LOOPRAW is larger than that shown in FIG. **5** because VFB remains below VREF for a longer period of time. The width of the negative pulse **610** is large enough so as to allow VCAP to ramp up to at least the threshold voltage of inverter **210** to thereby trigger a negative pulse for LOOPRAW_DLY. Starting at **631**, VCAP begins to ramp up at a first rate during **635** with switch SW1 being closed thereby bypassing resistor RF2. At **632**, the falling edge **621** of the LOOPRAW_DLY negative pulse **620** initiates a period of time in which LOOPRAW_DLY is logic low, which causes switch SW1 to turn off (open). With switch SW1 being off, charge current flows

through resistor RF2 as well as RF1. As a result of the extra resistance of RF2 created by opening switch SW1, the magnitude of the charge current to capacitor CF decreases and thus the rate at which capacitor CF charges decreases as shown at **637**. By decreasing the charge rate on capacitor CF, additional time is required for VCAP to reach TON_REF (compared to FIG. 5). When VCAP eventually reaches TON_REF at **640**, a falling edge **650** occurs on ONESHOT **157**. Logic **170** responds to the falling edge **611** of LOOPRAW **135** by turning on the HS transistor and responds to the falling edge **650** on ONESHOT **650** by turning off the HS transistor.

(28) The on-time of the HS transistor in the timing example of FIG. 6 has been extended and is greater than the on-time of FIG. 5. Accordingly, due to the larger output voltage undershoot, the width of the negative pulse on LOOPRAW increases, which provides enough time for LOOPRAW_DLY to be asserted low, which in turn causes the charge current to capacitor CF to decrease, thereby extending the time point at which ONESHOT **157** transitions to a logic low level.

(29) FIG. 7 illustrates a sudden increase in load current when VIN is near VOUT (e.g., Vin=6V and VOUT=5V). As shown in FIG. 7, the time period of each cycle is larger at **702** than at **701**, because that the on-time of the HS transistor has been extended during the undershoot portion **702**. At **720**, VCAP ramps up at faster rate than at **722**. The slower rate at **722** is due to switch SW1 being turned off to decrease the charge current to the capacitor CF as described above. The width of the negative pulses **710** are larger during the undershoot portion **702** compared to the narrow negative pulses **706** that occur with no significant output voltage undershoot. Advantageously, the peak undershoot **710** in FIG. 7 is shown as 480 mV (lower than the 740 mV of FIG. 4). Further, with HS transistor on-time being extended, the duty cycle of the HS transistor during the undershoot portion is larger (e.g., 95%) than the duty cycle of the HS transistor as illustrated in FIG. 4 (e.g., 88%).

(30) FIG. 8 shows a system **800** comprising a battery **805**, voltage regulator **100** (with on-time extension), and a load **810**. In one example, the system **800** is a notebook computer and the load includes one or more of the computer's processor, memory, and other electronics.

(31) The term “couple” is used throughout the specification. The term may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, in a first example device A is coupled to device B by direct connection, or in a second example device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B such that device B is controlled by device A via the control signal generated by device A.

(32) Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A voltage regulator circuit, comprising: a first transistor having a control input; a second transistor having a control input, the second transistor coupled to the first transistor at a switching terminal; a comparator having an output coupled to the control input of the first transistor, and having an input; a capacitor having a terminal coupled to the input of the comparator; and a first circuit capable of extending an on time for the first transistor in response a first feedback signal and a second feedback signal, wherein the first circuit includes a second circuit having an output coupled to the terminal of the capacitor and to the input of the comparator, the second circuit including a third transistor having a terminal and a fourth transistor having a terminal coupled to the terminal of the first transistor and to the output of the second circuit.
2. The voltage regulator circuit of claim 1, wherein: the first circuit is capable of controlling the on time of the first transistor based on a charge rate of the capacitor.
3. The voltage regulator circuit of claim 1, wherein: the first circuit includes: a first resistor coupled to the switching terminal; a second resistor coupled between the first resistor and the second input

of the comparator, the second resistor having first and second resistor terminals; a switch including first and second switch terminals and a control input, the first switch terminal is coupled to the first resistor terminal, and the second switch terminal is coupled to the second resistor terminal; and the output of the second circuit is coupled to the control input of the switch.

4. The voltage regulator circuit of claim 3, wherein: a charge current responsive to a voltage on the switching terminal is provided to the first circuit; and the capacitor is charged by the charge current.

5. The voltage regulator circuit of claim 3, wherein: the second circuit is a delay circuit having an input coupled to control inputs of the first and second transistors; and the delay circuit is capable of generating a falling edge on its output responsive to a signal on the input of the delay circuit being logic low for more than a delay time period after a falling edge of the signal on the input of the delay circuit.

6. The voltage regulator circuit of claim 5, further comprising: an inductor having a first terminal and a second terminal; and wherein: the first terminal of the inductor is coupled to the switching terminal; the first feedback signal is based on a voltage at the switching terminal; and the second feedback signal is based on a voltage at the second terminal of the inductor.

7. The voltage regulator circuit of claim 5, further comprising: a summing comparator having a first input, a second input, a third input, a fourth input, and an output coupled to the input of the delay circuit.

8. The voltage regulator circuit of claim 7, wherein: the first input of the summing comparator is configured to receive a feedback voltage; the second input of the summing comparator is configured to receive a voltage reference; the third input of the summing comparator is configured to receive a first loop compensation voltage; the fourth input of the summing comparator is configured to receive a second loop compensation voltage; and the output of the summing comparator is capable of generating the signal on the input of the delay circuit based on the first, second, third, and fourth inputs of the summing comparator.

9. A voltage regulator circuit, comprising: a first transistor having a control input; a second transistor having a control input, the second transistor coupled to the first transistor at a switching terminal; a logic circuit having a first and second control inputs and first and second outputs, the first output of the logic circuit is coupled to the control input of the first transistor and the second output of the logic circuit is coupled to the control input of the second transistor; and a one-shot circuit coupled to the logic circuit, wherein the one-shot circuit includes a delay circuit including a third transistor and a fourth transistor; wherein the one-shot circuit is capable of extending an on time for the first transistor in response to a delay signal based on a first feedback signal and a second feedback signal; and wherein the first feedback signal is based on a voltage of the switching terminal and the second feedback signal is based on an output voltage.

10. The voltage regulator circuit of claim 9, wherein: the logic circuit is capable of turning on the first transistor responsive to a signal edge on an output of the one-shot circuit.

11. The voltage regulator circuit of claim 9, wherein: the one-shot circuit includes a comparator; and the comparator has an output coupled to the first control input of the logic circuit.

12. The voltage regulator circuit of claim 11, wherein: the one-shot circuit is capable of generating a signal edge on its output at a time period after a signal edge on the output of the comparator, the time period defining an on-time for the logic circuit to maintain the first transistor in an on state; and the one-shot circuit is capable of lengthening the on-time responsive to a width of a pulse on the output of the comparator exceeding a threshold value.

13. The voltage regulator circuit of claim 9, wherein: the one-shot circuit includes: a first resistor coupled to the switching terminal; a capacitor; a second resistor coupled between the first resistor and the capacitor, the second resistor having first and second resistor terminals; a switch having first and second switch terminals and a control input, the first switch terminal is coupled to the first resistor terminal, and the second switch terminal is coupled to the second resistor terminal; and the

delay circuit having an input and an output, the output of the delay circuit is coupled to the control input of the switch.

14. The voltage regulator circuit of claim 13, wherein: in response to a voltage on the switching terminal, a charge current through the first resistor charges the capacitor; and a magnitude of the charge current is lower responsive to the switch being open than the switch being closed.

15. The voltage regulator circuit of claim 13, wherein: the input of the delay circuit is coupled to an output of a summing comparator.

16. The voltage regulator circuit of claim 15, wherein: the summing comparator has a first input, a second input, a third input, and a fourth input.

17. The voltage regulator circuit of claim 16, wherein: the first input of the summing comparator is configured to receive a feedback voltage; the second input of the summing comparator is configured to receive a voltage reference; the third input of the summing comparator is configured to receive a first loop compensation voltage; the fourth input of the summing comparator is configured to receive a second loop compensation voltage; and the output of the summing comparator is capable of generating a signal on the input of the delay circuit based on the first, second, third, and fourth inputs of the summing comparator.

18. The voltage regulator circuit of claim 13, wherein: the delay circuit is capable of generating a falling edge on the output of the delay circuit responsive to a signal on the input of the delay circuit being logic low for more than a delay time period after a falling edge of the signal on the input of the delay circuit.

19. The voltage regulator circuit of claim 18, wherein: a logic low on the output of the delay circuit causes the switch to be open; and a logic high on the output of the delay circuit causes the switch to be closed.

20. A system, comprising: a load; and a voltage regulator having an input voltage terminal and an output voltage terminal, the output voltage terminal coupled to the load; wherein the voltage regulator includes a first transistor coupled to a second transistor at a switching terminal, and the voltage regulator also includes a control circuit capable of: extending an on-time of the first transistor in response to a delay signal based on an amount of undershoot of an output voltage on the output voltage terminal being greater than a threshold; wherein the delay signal is further based on a voltage at the switching terminal; wherein the control circuit includes a one-shot circuit that includes a third and a fourth transistor; and wherein the one-shot circuit is capable of extending an on time for the first transistor in response to a delay signal based on a first feedback signal and a second feedback signal; and wherein the first feedback signal is based on a voltage of the switching terminal and the second feedback signal is based on an output voltage.
